Bank of Korea G5 (Statistics)

Practice problems for the written exam (major subject: statistics)

Part I: 40 practice problems Part II: 61 past exam problems (2019–2026 rounds, restated)

Solutions are in a separate booklet.

Notes. The exam consists of short-answer concept questions, derivations and computations in mathematical statistics, regression, sampling, design of experiments, time series and multivariate analysis, and (until the 2023 round) an essay on a statistical issue. Distribution tables are provided in the exam; critical values needed here are stated in the problems or solutions. z_α, $t_\alpha(\nu)$, $\chi^2_\alpha(\nu)$, $F_\alpha(\nu_1, \nu_2)$ denote upper-α points. Unless stated otherwise, round answers to two decimals.

Part I. Practice problems

A. Short-answer concepts

Problem 1

Concepts I

Briefly explain each of the following (three or four sentences each).

(a) The p-value of a test, and why it is not the probability that H_0 is true.

(b) Unbiasedness versus consistency of an estimator, with an example of an estimator that has one property but not the other.

(c) Simpson's paradox.

(d) Heteroskedasticity in a linear regression and its consequences for OLS.

(e) The (nonparametric) bootstrap.

Problem 2

Concepts II

Briefly explain:

(a) A sufficient statistic, and the factorization theorem.

(b) The power function of a test.

(c) Cointegration.

(d) The (augmented) Dickey–Fuller unit root test.

(e) Fixed-effects versus random-effects factors in an analysis of variance.

Problem 3

Concepts III (official statistics)

Briefly explain:

(a) Laspeyres and Paasche price indices, and the direction of their typical biases.

(b) Seasonal adjustment (e.g. X-13ARIMA-SEATS) and why seasonally adjusted figures get revised.

(c) The design effect of a complex survey.

(d) AIC and BIC.

(e) Overfitting and K-fold cross-validation.

B. Probability

Problem 4

Bayes's theorem

A bank's early-warning model flags firms as risky. Among firms that will default within a year (4% of all firms) the model flags 90%; among firms that will not default it flags 15%.

(a) What proportion of firms is flagged?

(b) What is the probability that a flagged firm defaults? That an unflagged firm defaults?

(c) Explain why the answer to (b) is low despite the high sensitivity, and how it would change if the model were applied only to a subpopulation with a 20% default rate.

Problem 5

Joint density

Let (X, Y) have density $f(x, y) = x + y$ on $0 < x < 1$, $0 < y < 1$.

(a) Find the marginal density of X and $E(X)$, $\mathrm{Var}(X)$.

(b) Find $\mathrm{Cov}(X, Y)$ and $\mathrm{Corr}(X, Y)$. Are X and Y independent?

(c) Find $P(X + Y < 1)$ and $E(Y \mid X = x)$.

Problem 6

Limit theorems

Claims arrive at an insurer at a rate of 100 per month, modelled as Poisson.

(a) Use Chebyshev's inequality to bound $P(|N - 100| \geq 20)$, where N is next month's count.

(b) Using the CLT with continuity correction, approximate $P(N \geq 120)$. (Exact value: 0.0282.)

(c) Prove the weak law of large numbers for i.i.d. variables with finite variance, and state the difference between the weak and strong laws.

(d) If $X_n \sim \text{Poisson}(n)$, find the limiting distribution of $2(\sqrt{X_n} - \sqrt{n})$ and explain why the square root is used.

Problem 7

Order statistics

Let $X_1, \ldots, X_n$ be i.i.d. Exp with mean θ (lifetimes of n independent components).

(a) Show that $X_{(1)} = \min X_i$ is exponential with mean θ/n, and hence that $nX_{(1)}$ is unbiased for θ. Compare its variance with that of $\bar{X}$.

(b) Using memorylessness, show $EX_{(n)} = \theta \sum_{k=1}^{n} \frac{1}{k}$.

(c) For a parallel system (works while at least one component works) of $n = 3$ components with $\theta = 2$ years, find the expected system life and $P(\text{system life} > 4)$.

C. Mathematical statistics

Problem 8

Estimation for the uniform distribution

Let $X_1, \dots, X_n$ be i.i.d. $U(0, \theta)$.

(a) Find the method-of-moments estimator $\tilde{\theta}$ and the MLE $\hat{\theta}$.

(b) Compute the bias and MSE of both. Find c so that $c\hat{\theta}$ is unbiased, and compare its MSE with that of $\tilde{\theta}$.

(c) Show that $X_{(n)}$ is complete and sufficient, and conclude which estimator is the UMVUE.

(d) Find the limiting distribution of $n(\theta - \hat{\theta})$. Why is the rate n rather than $\sqrt{n}$?

Problem 9

Neyman–Pearson and sample size

$X_1, \dots, X_{25}$ are i.i.d. $N(\mu, 10^2)$. Consider $H_0 : \mu = 100$ versus $H_1 : \mu = 104$ at $\alpha = 0.05$.

(a) Use the Neyman–Pearson lemma to derive the most powerful test.

(b) Compute its power. Is the test UMP for $H_1 : \mu > 100$? For $H_1 : \mu \neq 100$?

(c) How large must n be for power 0.90 at $\mu = 104$?

Problem 10

Likelihood ratio test for two exponential means

Independent samples $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Exp}(\text{mean } \theta_1)$ and $Y_1, \dots, Y_m \overset{\text{iid}}{\sim} \text{Exp}(\text{mean } \theta_2)$ are observed.

(a) Derive the LRT statistic for $H_0 : \theta_1 = \theta_2$ and show it depends on the data only through $T = \bar{X}/\bar{Y}$.

(b) Show that $T \sim F(2n, 2m)$ under H_0.

(c) With $n = m = 10$, $\bar{x} = 5$, $\bar{y} = 8$, carry out the test at level 0.05 (use $F_{0.025}(20, 20) = 2.46$).

Problem 11

Bayesian estimation

Monthly counts of operational incidents $X_1, \dots, X_n$ are i.i.d. Poisson(λ). The prior is $\lambda \sim$ Gamma(α, β) with density $\propto \lambda^{\alpha-1} e^{-\beta\lambda}$.

(a) Derive the posterior distribution and show that the posterior mean is a weighted average of the prior mean and the MLE.

(b) With $\alpha = 4$, $\beta = 2$, $n = 10$ and $\sum x_i = 31$, compute the posterior mean and a 95% equal-tailed credible interval (posterior quantiles: 2.03 and 3.96).

(c) Under absolute-error loss, what is the Bayes estimator? Contrast a credible interval with a confidence interval.

Problem 12

Rao–Blackwell and UMVUE

Let $X_1, \dots, X_n$ be i.i.d. Poisson(λ), $T = \sum X_i$, and $\eta = P(X = 0) = e^{-\lambda}$.

(a) Show that T is complete and sufficient.

(b) Starting from the unbiased estimator $\mathbf{1}\{X_1 = 0\}$, find the UMVUE of η.

(c) Compare it with the MLE $e^{-\bar{X}}$ for large n.

Problem 13

Fisher information and the delta method

Let $X_1, \dots, X_n$ be i.i.d. Bernoulli(p).

(a) Find the Fisher information $I(p)$ and the Cramér–Rao lower bound for unbiased estimators of p. Does $\bar{X}$ attain it?

(b) Find the asymptotic distribution of the estimated log-odds $\log \frac{\bar{X}}{1-\bar{X}}$.

(c) Construct an approximate 95% confidence interval for the log-odds and transform it into one for p, with $n = 200$ and $\bar{x} = 0.3$.

D. Regression

Problem 14

Simple linear regression from summary statistics

For $n = 12$ observations, $\bar{x} = 7.5$, $\bar{y} = 20$, $S_{xx} = 143$, $S_{xy} = 286$, $S_{yy} = 800$. Fit $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, $\epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$.

(a) Find $\hat{\beta}_0$, $\hat{\beta}_1$, the ANOVA table and R^2.

(b) Test $H_0 : \beta_1 = 0$ at $\alpha = 0.05$ by a t test and by an F test, and verify $F = t^2$. ($t_{0.025}(10) = 2.228$, $F_{0.05}(1, 10) = 4.96$.)

(c) Give a 95% confidence interval for β_1, for the mean response at $x_0 = 15$, and a 95% prediction interval for a new response at $x_0 = 15$. Explain why the last is wider.

Problem 15

The hat matrix and diagnostics

In $\mathbf{y} = X\beta + \epsilon$ with X of full column rank p, let $H = X(X'X)^{-1}X'$ and $\mathbf{e} = (I - H)\mathbf{y}$.

(a) Show that H and $I - H$ are symmetric and idempotent, and $\operatorname{tr} H = p$.

(b) Show $E\mathbf{e} = \mathbf{0}$, $\operatorname{Var}\mathbf{e} = \sigma^2(I - H)$, and $E(\mathbf{e}'\mathbf{e}) = (n - p)\sigma^2$.

(c) Define leverage h_{ii}, studentized residuals and Cook's distance, and explain what each detects.

Problem 16

Multicollinearity and omitted variables

(a) Show that $\mathrm{Var}(\hat{\beta}_j) = \dfrac{\sigma^2}{S_{jj}} \cdot \dfrac{1}{1 - R_j^2}$, where $S_{jj} = \sum_i (x_{ij} - \bar{x}_j)^2$ and R_j^2 is the R^2 from regressing x_j on the other regressors; $\mathrm{VIF}_j = 1/(1 - R_j^2)$. Compute VIF when $R_j^2 = 0.9$ and interpret.

(b) The true model is $y = \beta_1 x_1 + \beta_2 x_2 + \epsilon$ but y is regressed on x_1 only. Show $E\hat{\beta}_1 = \beta_1 + \beta_2 \delta$ where δ is the slope of x_2 on x_1. When is there no bias?

(c) List three remedies for multicollinearity and their costs.

Problem 17

Heteroskedasticity and weighted least squares

Consider $y_i = \beta x_i + \epsilon_i$ with independent errors, $E\epsilon_i = 0$ and $\operatorname{Var}\epsilon_i = \sigma^2 x_i^2$ $(x_i > 0)$.

(a) Show that the OLS estimator $\hat{\beta}_{\text{OLS}} = \sum x_i y_i / \sum x_i^2$ is unbiased and find its variance.

(b) Derive the WLS (GLS) estimator and show it equals the average of the ratios y_i/x_i. Show that it has smaller variance than OLS.

(c) Compute both for $(x, y) = (1, 2.2), (2, 3.9), (3, 6.4), (4, 7.6)$.

(d) In general, with $\operatorname{Var}\epsilon = \sigma^2 V$, state the GLS estimator and the Gauss–Markov result it satisfies.

Problem 18

Logistic regression

A logistic model for loan default ($y = 1$) on 100 borrowers uses debt-to-income ratio x_1 (%) and a dummy x_2 for self-employment: $\operatorname{logit} p = \beta_0 + \beta_1 x_1 + \beta_2 x_2$. The fitted coefficients are $\hat\beta_1 = 0.047$ (se $= 0.015$), $\hat\beta_2 = 0.47$ (se $= 0.30$). The null deviance is 120.4 (df 99) and the residual deviance 101.2 (df 97).

(a) Interpret $e^{\hat\beta_2}$ and $e^{10\hat\beta_1}$.

(b) Test $H_0 : \beta_1 = \beta_2 = 0$ by a likelihood-ratio test at $\alpha = 0.05$ ($\chi^2_{0.05}(2) = 5.99$).

(c) Test $\beta_2 = 0$ by a Wald test.

(d) Why is OLS on the 0/1 response (the linear probability model) problematic?

Problem 19

Structural change: Chow test

A consumption function $c_t = \beta_0 + \beta_1 y_t + \beta_2 r_t + \epsilon_t$ ($k = 3$ parameters) is fitted to quarterly data before ($n_1 = 20$) and after ($n_2 = 25$) a financial crisis. The pooled regression gives $SSE = 480$; the separate regressions give $SSE_1 = 170$ and $SSE_2 = 200$.

(a) Test at the 5% level whether the coefficients changed ($F_{0.05}(3, 39) = 2.85$).

(b) Show how the same test can be run as a single regression with dummy variables.

(c) What assumption about the error variances does the test require, and how would you check it?

E. Sampling

Problem 20

Sample size for a proportion

A survey will estimate the proportion of households holding crypto-assets.

(a) Under SRS from a large population, what sample size gives a 95% margin of error of 3 percentage points whatever the true proportion?

(b) Repeat for a population of $N = 20{,}000$ households.

(c) If the design is a cluster sample with 20 households per cluster and intraclass correlation 0.05, how many households are needed?

Problem 21

Ratio estimation

From $N = 500$ firms an SRS of $n = 25$ gives sales this year y and last year x: $\bar{x} = 40$, $\bar{y} = 52$, $s_x = 10$, $s_y = 15$, $s_{xy} = 120$. The population mean of last year's sales is known: $\bar{X} = 42$.

(a) Compute the ratio estimate of $\bar{Y}$ and its estimated variance.

(b) Compare with the SRS mean $\bar{y}$.

(c) Show that, to first order, the ratio estimator beats $\bar{y}$ iff $\rho > \frac{1}{2}\frac{C_x}{C_y}$, and check this condition here.

Problem 22

Stratified sampling and allocation

A population of $N = 10{,}000$ firms is split into three size strata with $N_h = (4000, 3000, 3000)$, means $(50, 80, 120)$ and standard deviations $S_h = (10, 20, 40)$. A sample of $n = 200$ is to be drawn.

(a) Find the variance of $\bar{y}_{st}$ under proportional allocation.

(b) Find the Neyman allocation and the resulting variance.

(c) Approximate the variance of the SRS mean and summarize the gains.

Problem 23

Cluster sampling

(a) In one-stage cluster sampling with M elements per cluster, show that $\mathrm{Var}(\bar{y}_{cl}) \approx \mathrm{Var}(\bar{y}_{\mathrm{SRS}})\{1 + (M-1)\rho\}$ where ρ is the intraclass correlation.

(b) Compare cluster and stratified sampling: how should groups be formed in each, and why?

(c) A survey samples 20 schools and all 30 pupils in each; $\rho = 0.1$. What is the effective sample size?

F. Design of experiments

Problem 24

One-way ANOVA with unequal replication

Three training programmes are compared by test scores:

A_1	18	21	20	23	
A_2	25	27	24	26	28
A_3	19	22	21		

(a) State the model and assumptions, and complete the ANOVA table. Test equality of means at $\alpha = 0.05$ ($F_{0.05}(2, 9) = 4.26$).

(b) Give a 95% CI for $\mu_2 - \mu_1$ ($t_{0.025}(9) = 2.262$). Why should multiple pairwise comparisons be adjusted (e.g. Tukey, Bonferroni)?

(c) How would you check the assumptions?

Problem 25

Randomized complete block design

Three fertilizers (A_1, A_2, A_3) are tested on four fields (blocks), each field divided into three plots assigned at random:

	B_1	B_2	B_3	B_4
A_1	12	15	14	17
A_2	14	18	17	19
A_3	10	12	13	14

(a) Write the model and complete the ANOVA table; test the fertilizer effect at $\alpha = 0.05$ ($F_{0.05}(2, 6) = 5.14$).

(b) Had the same data come from a completely randomized design, the block sum of squares would have been part of the error. Compute the error mean square in that case and the relative efficiency of blocking.

(c) Why can the treatment-by-block interaction not be tested here?

Problem 26

2^2 *factorial design*

Temperature (A) and catalyst (B), each at two levels, are studied with $r = 2$ replicates. The treatment totals are $(1) = 20$, $a = 30$, $b = 24$, $ab = 40$.

(a) Estimate the main effects and the interaction using contrasts, and compute their sums of squares.

(b) Given $SS_E = 6$ (4 df), test the three effects at $\alpha = 0.05$ ($F_{0.05}(1, 4) = 7.71$).

(c) Explain the meaning of a significant interaction and why main effects alone can mislead then.

Problem 27

Random-effects model

Five production batches are chosen at random and four units from each are measured. The ANOVA gives $MS_A = 60$ (df 4) and $MS_E = 12$ (df 15).

(a) Write the random-effects model and the expected mean squares.

(b) Test $H_0 : \sigma_\alpha^2 = 0$ ($F_{0.05}(4, 15) = 3.06$) and estimate the variance components and the intraclass correlation.

(c) What is the variance of the overall mean of the 20 measurements?

G. Time series

Problem 28

AR(1): moments and forecasting

$X_t - \mu = \phi(X_{t-1} - \mu) + \epsilon_t$, $\epsilon_t \overset{\text{iid}}{\sim} N(0, \sigma^2)$, with $\mu = 2$, $\phi = 0.6$, $\sigma^2 = 1$.

(a) Find the mean, variance and ACF ρ_k, and the PACF.

(b) Given $X_T = 5$, find the 1- and 2-step forecasts and 95% prediction intervals. What happens as the horizon grows?

(c) Explain how the Yule–Walker estimator of ϕ is obtained.

Problem 29

MA(1) and ARMA(1,1)

(a) For $X_t = \epsilon_t - \theta\epsilon_{t-1}$, find ρ_k. Show that θ and $1/\theta$ give the same ACF, and explain why the invertible choice $|\theta| < 1$ is used. Compute ρ_1 for $\theta = 0.5$.

(b) Show that $|\rho_1| \le \frac{1}{2}$ for every MA(1).

(c) For the ARMA(1,1) $X_t = \phi X_{t-1} + \epsilon_t - \theta\epsilon_{t-1}$ with $|\phi| < 1$, find the ψ-weights of its MA(∞) representation.

(d) Describe the ACF/PACF patterns used to identify AR(p), MA(q) and ARMA models.

Problem 30

Unit roots and spurious regression

(a) For the random walk with drift $y_t = \delta + y_{t-1} + \epsilon_t$, $y_0 = 0$, find Ey_t and $\operatorname{Var} y_t$. Why is it not stationary, and how is it made stationary?

(b) Two independent random walks are regressed on each other by OLS. Describe what typically happens to R^2, the t statistic and the Durbin–Watson statistic (spurious regression).

(c) Contrast trend-stationary and difference-stationary series and the consequences of detrending versus differencing each.

Problem 31

VAR and Granger causality

Consider the bivariate VAR(1) $\mathbf{y}_t = A\mathbf{y}_{t-1} + \mathbf{u}_t$ with $A = \begin{pmatrix} 0.5 & 0.2 \\ 0.3 & 0.4 \end{pmatrix}$, $\mathbf{y}_t = (\text{inflation}_t, \text{output gap}_t)'$.

(a) Is the process stationary?

(b) Compute the response of $\mathbf{y}_{t+1}$ and $\mathbf{y}_{t+2}$ to a unit shock in the output-gap equation at time t (non-orthogonalized).

(c) Define Granger causality and state the hypothesis that "output gap does not Granger-cause inflation" here.

(d) Why are orthogonalized impulse responses (e.g. Cholesky) used, and what is their drawback?

Problem 32

Exponential smoothing and seasonality

(a) Simple exponential smoothing forecasts $\hat{y}_{t+1} = \alpha y_t + (1-\alpha)\hat{y}_t$. Show that it is optimal for the ARIMA(0,1,1) model $\Delta y_t = \epsilon_t - \theta\epsilon_{t-1}$ with $\alpha = 1-\theta$.

(b) Write the seasonal ARIMA$(0,1,1)\times(0,1,1)_{12}$ (airline) model and explain each part.

(c) For monthly data, what do the seasonal difference $\Delta_{12}y_t$ and the year-on-year growth rate have in common?

Problem 33

GARCH and volatility

Daily returns follow $r_t = \sigma_t \eta_t$, $\eta_t \overset{\text{iid}}{\sim} N(0,1)$, $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$ with $\omega = 0.02$, $\alpha = 0.08$, $\beta = 0.9$.

(a) Find the unconditional variance and explain volatility persistence.

(b) Show that r_t is uncorrelated but not independent.

(c) Show that the kurtosis of r_t exceeds 3 for any ARCH(1) $\sigma_t^2 = \omega + \alpha r_{t-1}^2$ with $3\alpha^2 < 1$.

H. Multivariate analysis and computing

Problem 34

Principal components

$\mathbf{X} = (X_1, X_2, X_3)'$ has covariance matrix $\Sigma = \begin{pmatrix} 4 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

(a) Find the principal components and the proportion of variance each explains.

(b) Find the PCs of the correlation matrix and comment on the difference.

(c) How would you choose the number of components in practice?

Problem 35

Linear discriminant analysis

Two groups have $\mathbf{X} \sim N_2(\mu_k, \Sigma)$ with $\mu_1 = (0,0)'$, $\mu_2 = (2,1)'$, $\Sigma = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$, equal priors and costs.

(a) Derive the Bayes classification rule and show that it is linear.

(b) Compute the discriminant direction, the Mahalanobis distance Δ and the misclassification probability.

(c) How does the rule change with unequal priors π_1, π_2, or with unequal covariance matrices?

Problem 36

Monte Carlo integration and Newton–Raphson

(a) To estimate $\theta = \int_0^1 e^u\,du$, use $\hat\theta = \frac{1}{n}\sum e^{U_i}$ with $U_i \overset{\text{iid}}{\sim} U(0,1)$. Find $\operatorname{Var}\hat\theta$ and the n needed for a standard error of 0.001.

(b) Show that the antithetic estimator $\frac{1}{n}\sum_{i=1}^{n/2}(e^{U_i} + e^{1-U_i})$ has much smaller variance, and explain why.

(c) Write the Newton–Raphson iteration for the MLE of θ in a one-parameter model, and explain its relation to Fisher scoring.

Problem 37

Statistical learning

(a) State the bias–variance decomposition of the expected prediction error at a point x_0.

(b) Write the ridge and lasso criteria. Why does the lasso perform variable selection while ridge does not?

(c) For a classifier evaluated on 1000 cases with TP = 80, FP = 120, FN = 20, TN = 780, compute accuracy, precision, recall and specificity. Why can accuracy be misleading here?

I. Essay topics (Part II type)

Problem 38

Essay: alternative data and nowcasting

Card-transaction, web-search and mobile data are increasingly used to nowcast economic activity before official statistics are released. Discuss the statistical benefits and risks of using such data at a central bank, with concrete examples and methods.

Problem 39

Essay: revisions of official statistics

Preliminary GDP and seasonally adjusted figures are often revised substantially. Explain the sources of revisions and discuss how a statistical agency should balance timeliness and accuracy and communicate revisions to users.

Problem 40

Essay: survey nonresponse

Response rates to household surveys have been falling. Explain how nonresponse can bias survey estimates, and discuss design-stage and estimation-stage remedies.

Part II. Past exam problems (restated)

Problems are labelled by the year of the recruitment round (the exam is held in the preceding autumn). Statements are translated and occasionally condensed; data are as given in the exam.

2026 round (exam of 13 September 2025)

Past problem 1 *[2026 #1]*

Briefly describe: (a) type I error; (b) a $100(1-\alpha)\%$ confidence interval; (c) seasonality; (d) Monte Carlo integration; (e) principal component analysis.

Past problem 2 *[2026 #2]*

To survey monthly household water use, the country is divided into 3 regions and a stratified random sample is taken. Summary: $n_h = (6, 10, 8)$, $\bar{y}_h = (17.3, 33.2, 55.6)$, $s_h^2 = (21.6, 32.0, 85.6)$, $N_h = (200, 100, 100)$. (a) Give a 95% CI for the mean water use of all households in region 3 (take the bound as twice the estimated standard error). (b) Find $\bar{y}_{st}$ and $\hat{V}(\bar{y}_{st})$ for the whole country. (c) With known $\sigma_h = (5, 5, 10)$, costs per household $c_h = (16, 4, 25)$ and budget $\sum n_h c_h \le 900$, find the total n and the allocation n_h minimizing $V(\bar{y}_{st})$.

Past problem 3 *[2026 #3]*

In $y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i$, $\epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$, let R^2 be the coefficient of determination of the LS fit, and consider $H_0 : \beta_1 = \cdots = \beta_p = 0$. (a) Show that under H_0, $R^2 \sim \text{Beta}(\frac{p}{2}, \frac{n-p-1}{2})$. (b) Show that $W \sim \text{Beta}(a, b)$ implies $\frac{W/a}{(1-W)/b} \sim F(2a, 2b)$, and use this to derive the overall F test. (c) Show that $nR^2 \xrightarrow{d} \chi^2(p)$ under H_0.

Past problem 4 *[2026 #4]*

(a) For a random sample from $N(\mu, \sigma^2)$, show that $\bar{X}$ and $T = (X_1 - \bar{X}, \ldots, X_n - \bar{X})'$ are independent (use mgfs). (b) For a random sample $Y_1, \ldots, Y_n$ from $N(\theta, 1)$, show that $\bar{Y}$ is complete and sufficient. (c) Using (a), (b) and the unbiased estimator $\mathbf{1}_{(a,\infty)}(Y_1)$, find the UMVUE of $\eta = P_\theta(Y_1 > a)$.

Past problem 5 *[2026 #5]*

Substrate contamination is measured under 4 nozzle pressures (A) and 3 belt speeds (B), one observation per cell:

	B_1	B_2	B_3
A_1	1	4	3
A_2	1	3	5
A_3	4	5	7
A_4	5	4	6

(a) State the model and assumptions and complete the ANOVA table. (b) Test both factors at 5%. (c) Suppose the A_3B_2 value is missing (y). Explain how to estimate it by minimizing S_E (the answer is 5.67). (d) With the estimate, $S_T = 37.75$, $S_A = 18.56$, $S_B = 12.57$. Test both factors again and compare with (b).

Past problem 6 *[2026 #6]*

$X_1, \dots, X_n$ are independent $N(\alpha, \alpha\beta)$, $\alpha\beta > 0$. (a) Show that the MLE of (α, β) is $(\bar{X}, \hat{\sigma}^2/\bar{X})$ with $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$. (b) Find the MLE $\hat{\alpha}_R$ when $\beta = 1$. (c) Show that the LRT statistic for $H_0 : \beta = 1$ is $\lambda(\mathbf{x}) = \left(\frac{\hat{\sigma}^2}{\hat{\alpha}_R}\right)^{n/2} \exp\left\{\frac{n}{2} - \frac{\sum(x_i - \hat{\alpha}_R)^2}{2\hat{\alpha}_R}\right\}$.

Past problem 7 *[2026 #7]*

$y_t = \phi y_{t-1} + \varepsilon_t$, $\varepsilon_t = \sigma_t \eta_t$, $\eta_t \overset{\text{iid}}{\sim} N(0,1)$, $\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$ with $\alpha_0 > 0$, $\alpha_1 > 0$, $0 < \beta_1 < 1$, $\alpha_1 + \beta_1 < 1$. (a) Show that GARCH(1,1) is an ARCH(∞) and state its advantage over ARCH. (b) Find the unconditional variance of ε_t. (c) Show that ε_t^2 is ARMA(1,1), with $\nu_t = \varepsilon_t^2 - \sigma_t^2$.

Past problem 8 *[2026 #8]*

Twenty independent pairs (x_i, y_i) satisfy $x_1 < \cdots < x_{10} < 0 < x_{11} < \cdots < x_{20}$, $y_i = 0$ for $i \le 10$ and $y_i = 1$ for $i \ge 11$. Consider the logistic model $p_i = e^{\beta_0+\beta_1 x_i}/(1 + e^{\beta_0+\beta_1 x_i})$. (a) Show $\partial\ell/\partial\beta_0 = \sum(y_i - p_i)$ and $\partial\ell/\partial\beta_1 = \sum x_i(y_i - p_i)$. (b) Show $\ell(\beta_0, \beta_1) < 0$ for all (β_0, β_1). (c) Show that the MLE does not exist.

2025 round (exam of 28 September 2024)

Past problem 9 *[2025 #1]*

Briefly describe: (a) order statistics; (b) the most powerful test of simple $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$ at level α; (c) multicollinearity; (d) Neyman allocation in stratified sampling; (e) additive (AO) and innovational (IO) outliers in a stationary AR process.

Past problem 10 *[2025 #2]*

Compressive strength is measured for gypsum type A (2 levels) and gypsum amount B (3 levels), with 2 replicates: A_1: $(1,5)$, $(2,2)$, $(10,12)$; A_2: $(5,5)$, $(10,6)$, $(1,1)$ for B_1, B_2, B_3. (a) Write the model and test for interaction at $\alpha = 0.05$. (b) For the combination A_iB_j with the highest strength, give a 95% CI for its mean. (c) To find the best combination, is it better to estimate the factor means separately or the cell mean A_iB_j? Explain.

Past problem 11 *[2025 #3]*

$X_1, \ldots, X_n$ are independent with density $f(x) = 2^\theta \theta x^{-\theta-1}$, $x \geq 2$. (a) Find the MLE $\hat{\theta}$ and, using Fisher information, the asymptotic distribution of $\sqrt{n}(\hat{\theta} - \theta)$. (b) For $g(\theta) = \theta/(\theta - 1)$, find the asymptotic distribution of $\sqrt{n}\{g(\hat{\theta}) - g(\theta)\}$. (c) If $g'(\theta) = 0$ and $g''(\theta) \neq 0$ exists, find the asymptotic distribution of $n\{g(\hat{\theta}) - g(\theta)\}$.

Past problem 12 *[2025 #4]*

(a) For a discrete X with mgf $M_X(t)$, show $E(X^n) = M_X^{(n)}(0)$. (b) Find the mgf of $\text{Bin}(n, p)$. (c) Use it to find $\text{Var}\, X$.

Past problem 13 *[2025 #5]*

$y_t = \alpha y_{t-1} + \epsilon_t$, $t = 1, \ldots, n$, $|\alpha| < 1$, $y_0 = 0$, $\epsilon_t \overset{\text{iid}}{\sim} N(0, \sigma^2)$. (a) Write the joint density $p(y_1, \ldots, y_n)$ using conditional densities. (b) Find the mean and variance of the unconditional distribution of y_t. (c) Give $p(y_t \mid y_{t-1})$. (d) Give the log-likelihood $\ell(\alpha, \sigma^2)$.

Past problem 14 *[2025 #6]*

In $\mathbf{Y} = X\beta + \epsilon$ (X is $n \times (p+1)$), let X^* be the standardized regressors so that $X^{*\prime}X^*$ is the correlation matrix. (a) Find the sum of all eigenvalues of the correlation matrix and explain how eigenvalues diagnose multicollinearity. (b) For principal component regression, show that $\mathbf{Y}^* = X^*\beta^* + \epsilon^*$ can be written $\mathbf{Y}^* = C\alpha + \epsilon^*$ with C the $n \times p$ matrix of principal components. (c) Derive the LS estimator $\hat{\alpha}$ and explain how $\hat{\beta} = (\hat{\beta}_0, \ldots, \hat{\beta}_p)'$ is obtained from it.

Past problem 15 *[2025 #7]*

$(1-\alpha L)X_t = a_t$, $(1-\beta L)Y_t = b_t$ with independent white noises $a_t \sim WN(0,\sigma_a^2)$, $b_t \sim WN(0,\sigma_b^2)$. (a) Show that X_t is stationary iff $|\alpha| < 1$. (b) Write $Z_t = X_t + Y_t$ as an ARIMA(p,d,q) and decide stationarity for (i) $\alpha = \beta = 0.4$, (ii) $\alpha = 0.4, \beta = 0.5$, (iii) $\alpha = 1, \beta = 0.5$. (Hint: MA(k_1)+MA(k_2) is MA$(\max(k_1,k_2))$.)

Past problem 16 *[2025 #8]*

Sentiment $s_t = \alpha s_{t-1} + \varepsilon_t$, $\varepsilon_t \overset{\text{iid}}{\sim} N(0, \sigma_\varepsilon^2)$, $s_0 = 0$, is observed through the index $y_t = s_t + e_t$, $e_t \overset{\text{iid}}{\sim} N(0, \sigma_e^2)$, all shocks independent. (a) Find $E(s_t \mid s_0)$, $\mathrm{Var}(s_t \mid s_0)$ for $t = 1$ and $t > 1$. (b) Find $E(y_t \mid y_{t-1})$ and $\mathrm{Var}(y_t \mid y_{t-1})$ for $t = 1$ and $t > 1$. (c) With $\sigma_\varepsilon^2, \sigma_e^2$ known, derive the likelihood $L(\alpha; y_0, \ldots, y_T)$ (hint: Bayes's theorem).

2024 round (exam of 16 September 2023)

Past problem 17 *[2024 #1]*

Briefly answer: (a) census versus sample survey; (b) backward elimination in regression; (c) the odds of rolling a 1 with a fair die; (d) a complete statistic; (e) the central limit theorem.

Past problem 18 *[2024 #2]*

(a) Find k so that $P(X = x) = kx^3$, $x = 1, \dots, 5$, is a pmf; (b) find $P(1 \le X \le 2)$. The joint pmf of Y, Z is: $P(10, 10) = 0.3$, $P(10, 20) = 0.1$, $P(10, 30) = 0.2$, $P(20, 10) = 0.2$, $P(20, 20) = 0.1$, $P(20, 30) = 0.1$ (first argument Y). (c) Find the marginals. (d) Find EY, EZ and decide whether Y, Z are independent. (e) Find $E(Z \mid Y = 20)$.

Past problem 19 *[2024 #3]*

Prove: (a) for $X \sim U(a,b)$, $EX = \frac{a+b}{2}$ and $\operatorname{Var} X = \frac{(b-a)^2}{12}$; (b) as $n \to \infty$ with $np = \lambda$ fixed, the binomial converges to the Poisson.

Past problem 20 *[2024 #4]*

Let $\mathbf{i} = (1, \ldots, 1)'$ and $M = I_n - \mathbf{i}\mathbf{i}'/\mathbf{i}'\mathbf{i}$. (a) Show M is symmetric idempotent of rank $n-1$ and $M\mathbf{i} = 0$. (b) Show $\bar{y} = \mathbf{i}'\mathbf{y}/\mathbf{i}'\mathbf{i}$ and $s^2 = \mathbf{y}'M\mathbf{y}/(n-1)$. (c) If $\mathbf{y} \sim N(\mathbf{0}, \sigma^2 I_n)$, show $(n-1)s^2/\sigma^2 \sim \chi^2_{n-1}$. (d) Assuming $\bar{y}$ and s^2 independent, show $\bar{y}/\sqrt{s^2/n} \sim t_{n-1}$.

Past problem 21 *[2024 #5]*

(a) Y has density $24/y^4$, $y > 2$. For a sample of 48, approximate $P(\sum Y_i > 180)$ by the CLT. (b) $Z_1, \ldots, Z_n$ are i.i.d. $U(0, 2\theta)$. Find $EZ_{(n)}$ and an unbiased estimator of θ. (c) For $y_i = \beta x_i + \epsilon_i$ (no intercept, errors independent with mean 0, variance σ^2, independent of x), find the LS estimator $\hat{\beta}$, $E(\hat{\beta} \mid X)$ and $\mathrm{Var}(\hat{\beta} \mid X)$.

Past problem 22 *[2024 #6]*

The remaining life X of a bulb is exponential with density $\theta e^{-\theta x}$. (a) Show $\hat{\theta}_{\text{MLE}} = 1/\bar{X}$ and compute it for $x = (1, 2, 3)$. (b) An expert believes θ is exponential with mean $1/3$; give the prior. (c) Find the posterior for the data $(1, 2, 3)$. (d) Find the Bayes estimator (posterior mean) and contrast with the MLE.

Past problem 23 *[2024 #7]*

Battery life for exposure temperature A (3 levels, columns) and exposure time B (3 levels, rows), one run per cell in random order: B_1: $15, 10, 5$; B_2: $6, 13, 4$; B_3: $9, 8, 11$. (a) Write the model, conditions and hypotheses. (b) Fill in the decomposition $\sum\sum(x_{ij}-\bar{\bar{x}})^2 = \sum\sum(\ \ )^2 + \sum\sum(\ \ )^2 + \sum\sum(\ \ )^2$ and explain each term. (c)–(d) Complete the ANOVA table and test at 5%.

Past problem 24 *[2024 #8]*

To estimate per-capita income, 20 of 300 clusters are sampled ($\sum m_i = 129$, $\sum y_i = 11{,}990$; data in the exam). (a) Estimate per-capita income and its variance. (b) Treating this as a pilot, what number of clusters gives a bound of 50? (c) A neighbouring city (stratum 2) has 200 clusters, of which 10 are sampled ($\sum m_j = 43$, $\sum y_j = 3800$). Estimate the per-capita income of both cities together and its variance. (d) Name the design in (c) and compare (a) with (c).

Past problem 25 *[2024 #9]*

(a) What problem arises when estimating a relation between two nonstationary series, and how is it solved? (b) Explain how to test for cointegration and what to watch for. (c) $x_t, y_t \sim I(1)$ are cointegrated with $y_t = 5x_t + \epsilon_t$, $\epsilon_t \sim I(0)$, and $y_t = 4 + 0.5y_{t-1} + 2x_t + \theta x_{t-1} + u_t$ is estimated. Derive the error-correction model and find θ.

2023 round (exam of 24 September 2022)

Past problem 26 *[2023 #1]*

Briefly describe: (a) Spearman's rank correlation; (b) communality in factor analysis; (c) the Gauss–Markov theorem; (d) the variance inflation factor; (e) the ARCH model.

Past problem 27 *[2023 #2]*

$Y_i = \mu + u_i$, where u_i is 1 or 0 with probability $\frac{1}{2}$ each and $\mu \in \{0, 1\}$. With n observations test $H_0 : \mu = 0$ vs $H_1 : \mu = 1$. (a) Find a test with type I error 0 and positive power. (b) Find its power as a function of n.

Past problem 28 *[2023 #3]*

Fibre elasticity at four concentrations $(A_1, \dots, A_4)$, three runs each in random order: A_1: 12.0, 12.0, 11.0; A_2: 11.0, 9.0, 10.0; A_3: 7.0, 9.0, 9.0; A_4: 7.5, 8.5, 8.0. (a) Model, conditions, ANOVA table. (b) Test A at 5% and give a 90% CI for $\mu(A_3)$. (c) Give a 95% CI for σ_E^2. (d) In another experiment the four concentrations are run on 4 randomly chosen days B (16 runs). Describe the model and how it differs.

Past problem 29 *[2023 #4]*

A wholesaler estimates mean sales per outlet over 4 chains ($N_h = 24, 36, 30, 30$, $N = 120$) with a stratified sample of 20: chain A: 94, 90, 102, 110; B: 91, 99, 93, 105, 111, 101; C: 108, 96, 100, 93, 93; D: 92, 110, 94, 91, 113 ($s_h^2 = 78.67, 55.6, 39.5, 112.5$). (a) Find $\bar{y}_{st}$ and $\hat{V}(\bar{y}_{st})$. (b) Treating the data as an SRS, find $\bar{y}$ and $\hat{V}(\bar{y})$. (c) Compare.

Past problem 30 *[2023 #5]*

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$. (a) Find the method-of-moments estimators of μ, σ^2. (b) Find the MLEs. (c) Discuss their unbiasedness and consistency.

Past problem 31 *[2023 #6]*

$\dot{X}_t = 0.8\dot{X}_{t-1} - 0.1\dot{X}_{t-2} + \epsilon_t$, $\dot{X}_t = X_t - \mu$. (a) Is it stationary? If so, write it as an MA process. (b) Find ρ_1, ρ_2.

Past problem 32 *[2023 #7]*

Standardized regression $\mathbf{y} = X\beta + \epsilon$, $\operatorname{Var}\epsilon = \sigma^2 I$, with $X'X = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}$. (a) With $\sigma^2 = 1$, find $\operatorname{Var}\hat{\beta}^{\mathrm{OLS}}$. (b) Derive the ridge estimator minimizing $(\mathbf{y} - X\beta)'(\mathbf{y} - X\beta) + \lambda\beta'\beta$. (c) Is it unbiased? Find its bias. (d) Find its variance for $\lambda = 1/5$, compare with (a), and discuss.

Past problem 33 *[2023 #8]*

$\mathbf{X} = (X_1, X_2)' \sim N_2(\mu, \Sigma)$ with $\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$. (a) Write the density. (b) Show $X_2 \mid X_1 = x_1 \sim N(\mu_2 + \rho\sigma_2(x_1 - \mu_1)/\sigma_1,\ \sigma_2^2(1-\rho^2))$. (c) For $\mu = (3, 1)'$, $\Sigma = \begin{pmatrix} 16 & 12 \\ 12 & 25 \end{pmatrix}$, compute $P(5 < X_2 < 12 \mid X_1 = 7)$. (d) For $\mu = 0$, $\Sigma = I$, find $E\max(X_1, X_2)$.

Past problem 34 *[2023 #9 (essay)]*

As data grow, many new statistics are being developed to capture social and economic phenomena, but statistics without proven stability may confuse users. Using the example of the "lipstick index" (which failed during the COVID recession because of mask-wearing), discuss whether diversity or stability should take priority when a statistics producer publishes new statistics.

2022 round (exam of 11 September 2021)

Past problem 35 *[2022 #1]*

In $Y = X\beta + \epsilon$ (X is $n \times p$), $\epsilon \sim N(0, \sigma^2 I_n)$: (a) show $SSE = Y' M_X Y$ with $M_X = I - X(X'X)^{-1}X'$; (b) using the theorem "if $y \sim N(\mu, \Sigma)$ and $P\Sigma$ is idempotent then $y'Py \sim \chi^2(\text{rank}(P\Sigma), \frac{1}{2}\mu' P \mu)$", show $SSE/\sigma^2 \sim \chi^2_{n-p}$ and that MSE is unbiased; (c) for $\beta = (\beta_1', \beta_2')'$ with β_2 of dimension $p - r$, test $H_0 : \beta_2 = 0$ given $SSE_R = 672$ (df 22) and $SSE_F = 480$ (df 20), at $\alpha = 0.05$.

Past problem 36 *[2022 #2]*

(a) A die rolled 120 times gives counts $18, 23, 16, 21, 18, 24$. Test fairness at 5%. (b) A fair coin is tossed once and a fair die six times. The player wins if (heads and at least four of the six rolls are ≥ 5) or (tails and at least five rolls are < 5). Find the winning probability. (c) Prove that $\text{Bin}(n, \theta) \to \text{Poisson}(\lambda)$ as $n \to \infty$, $\theta \to 0$, $n\theta = \lambda$.

Past problem 37 *[2022 #3]*

$y_1, \dots, y_n$ is an SRS (without replacement) from $Y_1, \dots, Y_N$ with mean μ and variance $\sigma^2 = \frac{1}{N}\sum (Y_j - \mu)^2$. (a) Show $\operatorname{Var} \bar{y} = \frac{\sigma^2}{n}\frac{N-n}{N-1}$. (b) Show $\frac{N-n}{N}\frac{s^2}{n}$ is unbiased for $\operatorname{Var} \bar{y}$. (c) With $N = 3000$, $n = 300$, $\bar{y} = 320$, $s^2 = 1470$, estimate μ and the bound (twice the estimated se). (d) Using proportional stratified sampling by income quintile with $s_h^2 = 540, 1350, 1620, 1350, 540$, find the estimated standard error.

Past problem 38 *[2022 #4]*

For a Poisson(λ) sample: (a) show the family has monotone likelihood ratio; (b) define a UMP test and find the UMP test of $H_0 : \lambda \le \lambda_0$ vs $H_1 : \lambda > \lambda_0$.

Past problem 39 *[2022 #5]*

In $\mathbf{Y} = X\beta + \epsilon$: (a) if $\epsilon \sim N(0, \sigma^2 I)$, find the distribution of $\hat{\beta}_{\text{OLS}}$ and prove it is BLUE; (b) if $\epsilon \sim N(0, V)$ with $V = \text{diag}(\lambda_1, \ldots, \lambda_n) \neq \sigma^2 I$, is OLS still BLUE? If not, give the BLUE.

Past problem 40 *[2022 #6]*

(a) Fill in the Newton–Raphson pseudocode for the MLE: $x_1 = \mathbf{A}$; $i = \mathbf{B}$; if $|\ell'(x_1)| > e$ then $x_0 = \mathbf{C}$ and repeat. (b) Describe the bootstrap in about two sentences. (c) For $X \sim N(\mu, \Sigma)$ with $\Sigma = \Gamma' \Lambda \Gamma$, state the decomposition used and the properties of $Y = \Gamma(X - \mu)$. (d) Write pseudocode for k-means. (e) Describe nearest-neighbour interpolation and its drawback.

Past problem 41 *[2022 #7]*

$Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$ with prior $p \sim \text{Beta}(\alpha, \beta)$. (a) Find the posterior. (b) Find the posterior mean. (c) Show that under squared-error loss the Bayes estimator minimizing the posterior expected loss is the posterior mean.

Past problem 42 *[2022 #8]*

$X_1, \ldots, X_n$ i.i.d. with density $\frac{1}{2\theta}$ on $|x| < \theta$. (a) Find the MLE $\hat{\theta}_n$. (b) Find the cdf of $\theta - \hat{\theta}_n$. (c) Find C_n such that $C_n(\theta - \hat{\theta}_n)$ has a nondegenerate limit, and the limit.

Past problem 43 *[2022 #9]*

(a) Find the ACF of the stationary AR(1) $X_t = \delta + \phi X_{t-1} + \epsilon_t$. (b) Of the MA(1) $X_t = \epsilon_t - \theta\epsilon_{t-1}$. (c) Of the stationary ARMA(1,1) $X_t = \delta + \phi X_{t-1} + \epsilon_t - \theta\epsilon_{t-1}$. (d) Explain identification with ACF/PACF, its limitations, and model selection criteria.

Past problem 44 *[2022 #10 (essay)]*

Central banks are increasingly interested in machine learning (neural networks, data mining) for economic research, financial stability and monetary policy, while data quality, privacy and algorithmic fairness remain concerns. Discuss how traditional statistical modelling and machine learning should be used, with examples, and give your view.

2021 round (exam of 12 September 2020)

Past problem 45 *[2021 #1]*

The joint pmf of $X \in \{0,2,4\}$ and $Y \in \{1,2,3\}$ is (in thirtieths, rows $Y=1,2,3$): $(2,3,3)$, $(5,4,3)$, $(2,8,0)$. (a) Find the marginals. (b) Find EX, EX^2, $\operatorname{Var} X$, EY, $E(X+Y)$. (c) Are X, Y independent?

Past problem 46 *[2021 #2]*

(a) If $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$, show $X_n + Y_n \xrightarrow{p} X + Y$ and $aX_n \xrightarrow{p} aX$. (b) If $X_n \xrightarrow{p} a$ and g is continuous at a, show $g(X_n) \xrightarrow{p} g(a)$. (c) Show $\bar{X} \xrightarrow{p} \mu$ for i.i.d. variables with finite variance. (d) So $\bar{X}$ is a _____ estimator of μ.

Past problem 47 *[2021 #3]*

(a) For a continuous population with cdf F and $h(y) = F^{-1}(1 - e^{-y})$, show $(X_{(r)})_r \overset{d}{=} \big(h(\frac{Z_1}{n} + \cdots + \frac{Z_r}{n-r+1})\big)_r$ with $Z_r \overset{\text{iid}}{\sim} \text{Exp}(1)$. (b) Gini's mean difference is $G_n = \binom{n}{2}^{-1} \sum_{i<j} |X_i - X_j|$. For $n = 10$ find a_i with $G_{10} = \sum a_i X_{(i)}$. (c) Show $E|X - \mu| = \sigma\sqrt{2/\pi}$ for $X \sim N(\mu, \sigma^2)$. (d) Show $EG_n = 2\sigma/\sqrt{\pi}$ for normal samples.

Past problem 48 *[2021 #4]*

Cutting force of four tool types (new tool each run): ① $66, 56, 70$; ② $55, 57, 60$; ③ $45, 40, 51$; ④ $60, 56, 53$. (a) Model and constraints. (b) Decompose SS_T. (c) ANOVA and test at 5%. (d) 95% CIs for the means of ① and ③ and for their difference.

Past problem 49 *[2021 #5]*

$Z_t = \epsilon_t + 2.4\epsilon_{t-1} + 0.8\epsilon_{t-2}$, $\epsilon_t \sim WN(0,1)$. (a) Find the autocovariances. (b) Show Z_t is not invertible. (c) Find an invertible MA(2) with the same autocovariances.

Past problem 50 *[2021 #6]*

In $Y = X\beta + \epsilon$, $\epsilon \sim N(0, \sigma^2 I)$: (a) give the test statistic for $H_0 : C\beta = m$ (C is $q \times p$ of rank q); (b) show that for $H_0 : \beta_j = 0$ the t and F tests agree.

Past problem 51 *[2021 #7]*

Independent samples $X_1, \dots, X_n \sim N_p(\mu_x, \Sigma)$ and $Y_1, \dots, Y_m \sim N_p(\mu_y, \Sigma)$; pooled $S = \frac{(n-1)S_x + (m-1)S_y}{n+m-2}$. (a) Write the likelihood. (b) Find the MLEs of μ_0 and Σ when $\mu_x = \mu_y = \mu_0$. (c) Find the unrestricted MLEs.

Past problem 52 *[2021 #8 (essay)]*

With COVID-19 increasing uncertainty, demand has grown for statistics that assess the economy quickly and accurately. Propose responses (new statistics, improved methods), with examples, and evaluate them in terms of timeliness and accuracy.

2019 round (exam of 20 October 2018)

Past problem 53 *[2019 #1]*

Briefly describe: (a) coefficient of variation; (b) Jensen's inequality; (c) the logistic distribution; (d) the memoryless property of the exponential distribution; (e) Fisher information.

Past problem 54 *[2019 #2]*

For continuous X_1, X_2 with $\operatorname{Var} X_2 < \infty$, show (a) $E\{E(X_2 \mid X_1)\} = EX_2$; (b) $\operatorname{Var}\{E(X_2 \mid X_1)\} \leq \operatorname{Var} X_2$.

Past problem 55 *[2019 #3]*

When the population is divided into groups, compare cluster sampling and stratified sampling with regard to the characteristics of the groups.

Past problem 56 *[2019 #4]*

A simple regression on 10 observations gives $\hat{\beta}_0 = 3.5$ (se 1.12), $\hat{\beta}_1 = 8.4$ (se 2.34) and $SSE = 44.8$. (a) Give a 95% CI for β_1. (b) Fill in the ANOVA table (regression F, error df, total SS) and find R^2. (c) If $\beta_0 = 3.5$ is known, propose a model, derive the LS estimator and discuss.

Past problem 57 *[2019 #5]*

For the stationary AR(1) $X_t = \phi_0 + \phi_1 X_{t-1} + \epsilon_t$, $\epsilon_t \overset{\text{iid}}{\sim} N(0, \sigma^2)$: (a) find the method-of-moments estimators of ϕ_0, ϕ_1; (b) find the conditional MLEs given $X_1 = x_1$.

Past problem 58 *[2019 #6]*

$f(x) = (2 - 4|x - 0.5|)\mathbf{1}_{(0,1)}(x)$. (a) Find Ee^{tX}. (b) Find EX, check that it equals $\lim_{t\to 0} \frac{d}{dt} Ee^{tX}$, and discuss whether the mgf exists.

Past problem 59 *[2019 #7]*

$X_1, \ldots, X_n$ are i.i.d. with density $\alpha e^{-\alpha x}$, $x > 0$. (a) Find the density of $Y = \beta e^X$ ($\beta > 0$). (b) Find EY^n, EY, $\operatorname{Var} Y$. (c) Find the MLEs of α, β from $Y_1, \ldots, Y_n$. (d) Approximate the variance of the MLE in large samples.

Past problem 60 *[2019 #8]*

State and prove the Neyman–Pearson lemma.

Past problem 61 *[2019 #9 (essay)]*

Statistics are sometimes compiled from statistics produced by several other agencies (e.g. aggregating separately estimated sub-sectors, or combining several agencies' data to estimate a sector). Explain the problems that arise and their causes, and what the compiler should check or do.